Report

UNIVERSITY OF SKÖVDE

AI-POWERING THE MARKETING OF A STUDENT-LED NON-PROFIT ORGANISATION

How AI supported content marketing, advertising, and media buying can enhance member applications and audience engagement of KTH AI Society

Introduktion till AI-tillämpningar i marknadsföringsstrategi G1N

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Introduction

KTH AI Society (KTHAIS) is a student-led non-profit organisation at KTH Royal Institute of Technology, that aims to introduce and expand AI knowledge among students at KTH. This is pursued by writing AI-research and producing AI-developments, as well as organising AI-related events such as Hackathons, panels, lunch lectures, company visits, and other educational events (KTH AI Society, n.d.).

As a student-led non-profit organisation, KTHAIS differs from profit-maximising firms in its primary objectives, however, the direction is similar. For the organisations long-term sustainability depends on attracting and retaining members, maintaining high levels of student engagement, and securing external funding. Since it is student-led, both the executive board and active members change regularly due to annual elections and voluntary participation, thus needing the organisation to continuously recruit skilled and motivated students (KTH AI Society, 2024, sec. 4,6). At the same time, to keep get funding for research, development and events, the organisation must upkeep a good image and outreach to incentivise partnerships with sponsoring companies (Michel & Rieunier, 2011). Consequently, KTHAIS faces two primary marketing challenges: recruiting capable students as member while simultaneously keeping the interest of students, industry professionals, and corporate partners to keep events motivating and stimulating. These challenges are expected to be addressed through AI-driven marketing strategies.

The aim of this report is thus to design an AI-driven marketing strategy that strengthens KTHAIS outreach to both students, professionals and corporate partners. The strategy focuses on applying specific artificial intelligence technologies, including Machine Learning, Natural Language Processing, predictive analytics, and AI-supported advertising to improve customer segmentation, personalise communication, and optimise media buying. By leveraging behavioural data and continuous learning, the proposed strategy seeks to increase engagement, recruitment, and sponsorship opportunities while maintaining responsible AI practices and ensuring that marketing activities create measurable value for the organisation.

Background and Marketing Context

These two marketing challenges addresses two different targeted markets; students currently enrolled at KTH with an interest for artificial intelligence and actors with an interest and optimally an expertise in AI. At present, KTHAIS does not employ a specialised marketing strategy and primarily relies on organic outreach through platforms such as LinkedIn which also is not ad incentivised. It therefor remains unclear whether the limited engagement is primarily caused by insufficient awareness of the organisation or by a lack of interest of the audience. No matter the case, both factors are likely to contribute to the current challenge. Informal observations suggest that many students are unfamiliar with KTHAIS, while those who recognise the organisation often have limited understanding of it does other than highly visible events such as hackathons which are usually promoted through posters on campus. Although online engagement is more difficult to evaluate due to limited access to behavioural data, AI-based processes provide ways to better understand audience interactions and interests (Ahamed, 2026a).

The primary target market for KTHAIS is thus comprised of the student body at KTH, a population characterized by high technical literacy but varying levels of AI expertise. Within this market, the segmentation should be driven not just by program, but by their behavioural traces (Bano et al., 2025). These include student who frequently engage with AI related events, engage with AI-focused content, search for technical resources, or repeatedly visit the organisations event and project pages. Traditional student needs include professional networking and technical skill acquisition. However, AI-driven marketing reveals that these needs are anticipatory (Ahamed, 2026e). For example, students searching for specific programming libraries, academic papers, or AI-related tools may represent potential members even if they have not previously interacted with KTHAIS. Noteworthy is that the sheer volume of student chatter on platforms like Discord and LinkedIn makes manual assessment impractical.

The secondary target market consists of AI professionals and corporate partners within a B2B value co-creation ecosystem. Within this market, target customers are segmented not by broad categories, but by granular patterns of expertise and intent. This could be companies seeking increased brand visibility, access to technical talent, opportunities for recruitment, and collaboration through sponsored events, but it also includes AI-professionals pursuing advanced knowledge exchange or seeking to transition into AI. For these segments, digital footprints can also be analysed in a similar way as for the students, but with different parameters. However, compared with the student market, corporate partners and professionals place greater emphasis on trust, credibility, and responsible AI practices when evaluating potential collaborations.

Theoretical and Conceptual Framework

Artificial intelligence in marketing is defined as systems performing tasks associated with human intelligence, primarily used to classify, forecast, and optimise marketing actions (Ahamed, 2026c). Within this strategy, the focus is on AI-learning (Machine Learning, ML), Natural Language Processing (NLP) and predictive analytic. ML allows systems to independently identify patterns within data and adapt based on previous observations, thus improving their performance (Naqa & Murphy, 2015). Many data-handling approaches, like K-Means and XGBoost, are grounded in machine learning. Algorithms like K-Means find natural groups or clusters within data without predefined labels, learning from the data gathered over each iteration. This can be used for pattern discovery in students, professionals and company's interests (Valeur & Liekis, 2023). Algorithms like XGBoost use their repetitive feedback to learn from historical outcomes (such as whether a student applied in the past) to forecast future actions. This allows the organisation to move from describing who customers are to anticipating what they will do next (Bentéjac et al., 2019; Kumar & L, 2018). Prescriptive Analytics, which XGBoost is part of, is however often categorised separately from ML although it uses machine learning. NPL complements machine learning by connecting raw text (from places like discord chats, or LinkedIn posts) to structured indicators including sentiment, topics, and intent. It transforms unstructured discourse into usable marketing signals, thus turning language into assets (Nadkarni et al., 2011). Another relevant AI application is programmatic advertising, including real-time bidding (RTB). RTB systems automatically optimise advertisement placement by analysing user data and adjusting bidding strategies dynamically according to predicted engagement potential (Yuan et al., 2013; Zhang et al., 2014).

The proposed strategy is based on the analysis of digital traces such as search queries, website activity, user-generated content, and media interactions. This data it then processed by NPL's and ML's to created valuable internal signals. By combining these behavioural, contextual, and psychographic signals, AI can uncover micro-segments that traditional rules miss, leading to higher targeting precision (Ahamed, 2026e; Ahamed, 2026f; Bano et al., 2025). Personalized content with consistent publishing affects the business outcome of marketing significantly (Clary, 2026). This further motivates the reason to ensure that the perfect content contains the most usable information at their specific moment of need (Ahamed, 2026b).

AI-Driven Marketing Strategy or Campaign

The proposed marketing strategy is built on two elements, the use of AI for segmenting and finding the targeted market and using AI in advertising and media buying. AI will however not be used to create the marketed content. The brand of non-profit organisation correlates to about 30-40% of money bringing interactions (Michel & Rieunier, 2011). Research also emphasised the weight of having a consistent brand, especially for campaigns (Di Lauro et al., 2019; Michel & Rieunier, 2011). The present AI is still lacking regarding the ability to produce consistent content. Since KTHAIS already maintains established marketing guidelines and content structures, the strategy prioritises human-created content supported by AI-driven distribution and optimisation. In this approach, AI strengthens the reach and effectiveness of existing communication rather than replacing the organisation's creative processes (Li & Xiao, 2025; Ahamed, 2026d).

For segmentation, KTHAIS will shift from broad demographics toward an ongoing learning process that captures dynamic digital traces, such as clickstreams and sentiment data (Ahamed, 2026a; Ahamed, 2026g). This process will discover hidden customer patterns using a hybrid approach of unsupervised clustering (K-Means) for group discovery and supervised prediction (XGBoost) for estimating conversion probability. Natural Language Processing (NLP) and sentiment analysis will further support segmentation by transforming unstructured communication into measurable marketing signals. Data from sources such as professional discussions, social media interactions, and online communities can provide insights into topics of interest, attitudes towards AI, and potential engagement intentions (Haider et al., 2025). This enables KTHAIS to identify audiences with a high likelihood of engagement and adapt communication accordingly. While a multi-modal strategy (analysing text, images, and video) could increase accuracy, it is decided to use only text-handling models due to the high computational costs and resource constraints of complex transformer models (Rita et al., 2025). Additionally, the model will analyse corporate feedback to refine the "value proposition" of sponsorship packages, identifying specific "unmet needs" in the B2B sector (Haider et al., 2025).

Regarding AI advertising, most social-media (SM) platforms already support automated systems that manage the buying and placement of ads (Achnani, 2025; Anute et al., n.d.). Findings from the AI segmentation will be used to target specific users, followed by internal AI media buying to deliver ads at the optimal time and place (Anute et al., n.d.). Automated optimisation methods, such as Meta's Dynamic Creative Optimization will be utilized to automatically test headlines and images to see which version resonates best with partners, thereby increasing click-through rates (Singh, 2026). Furthermore, programmatic media buying and RTB will support through more efficient advertisement placement by dynamically adjusting advertising decisions based on predicted performance (Anute et al., n.d.). These platforms use recommendation engines to suggest research papers or events based on a visitor's browsing history, acting at moments when they are most susceptible to influence. For instance, a student finishing a "Python for Data Science" course may receive a "Next Best Offer," such as an invitation to a beginner-friendly hackathon (Ahamed, 2026f). For companies, the strategy enables KTHAIS to provide quantifiable insights into the student talent pool, transforming sponsorship from a donation into a strategic investment (Anute et al., n.d.; Haider et al., 2025).

The model and sediment analysis will be set to flag AI-related searches and high-level programming interactions as indicators of intent. From this, AI is implemented to forecast conversion probability, the likelihood of a student applying or a company seeking partnership, as well as the economic strength and "propensity to purchase" of corporate partners. To keep costs down, the society will not utilize all platforms, limiting outreach to Facebook, LinkedIn, Professional Forums, the KTHAIS Website, and Email Marketing, relying on simple browsing information gathered from cookies to maintain a data foundation. This approach ensures KTHAIS can recognize customer journeys while maintaining their authenticity and credibility.

Implementation and Evaluation

The success of the strategy will primarily be evaluated through engagement rate, click-through rate, and revenue generated though posts, as these metrics directly reflect the strategic objectives of the strategy. Reach, defined as the total number of unique users who see the recruitment or partnership content, will also be monitored to place engagement into perspective. Measuring reach enables engagement metrics, such as likes, comments, and shares, to be interpreted relative to audience size, allowing KTHAIS to identify which types of content generate the strongest response and continuously optimise future campaigns. However, reach itself will not constitute a primary performance indicator. The click-through rate will provide an indicator of campaign success by measuring the proportion of users who proceed from content interaction to further action. As campaign links are expected to direct users to event registrations, membership applications, or partnership information, a higher click-through rate represents increased user interest and progression along the intended customer journey. However, while engagement and click-through behaviour indicate marketing effectiveness, they do not directly reflect organisational sustainability. Consequently, the total revenue generated through sponsorships, event registrations, or other campaign-related activities will serve as the final performance metric, demonstrating whether marketing activities translate into tangible organisational value.

A/B testing is performed inherently in AI-driven advertising and media buying. This is thus not something the strategy itself will implement. As AI-supported advertising is a central component of the strategy, conversion lifts will be executed regularly to ensure that identified segments remain meaningful. Silhouette testing will be used to evaluate the cohesion and separation of the identified clusters, thus assessing the effectiveness of the customer segmentation. A feature importance analysis would also be presumed to be in place, but as this would happen more on the technical level than the marketing level, it is not part of the strategy. The same is true for data lineage tracking and anomaly correction targets. On similar notes, it is also assumed that on a technical level, metrics like precision, recall, and F1-scores are used to evaluate the model, thus constantly bettering the model. The implementation of AI-driven marketing should therefore be viewed as an iterative process rather than a one-time deployment. By continuously analysing campaign results, audience responses, and model performance, KTHAIS can refine its marketing activities and ensure that AI applications remain aligned with organisational objectives.

Ethical Considerations and Responsible AI Use

The strategy's reliance on granular behavioural traces necessitates a robust governance framework. For personalization to be effective and sustainable, it must be balanced with transparency and fairness. A central consideration is the collection and processing of behavioural data. The information proposed to gather contains website interactions, search behaviour, social media activity, and engagement patterns that may reveal insights into individual interests and intentions, however, consumers should understand how their social and behavioural data influences marketing decisions. The use of such data must thus follow the principles of data minimisation, purpose limitation, and transparency which also are established by GDPR and the AI Act. As per these laws, only data necessary for achieving defined marketing objectives will be collected, and unnecessary exposure of personal information should be avoided.

Another important consideration is algorithmic fairness. Since predictive models learn from historical data, the existing inequalities or biases within their training-data may be reproduced or amplified. For example, if previous engagement data disproportionately represents certain study programmes, the AI model may unintentionally prioritise this group while reducing visibility among underrepresented audiences. Continuous monitoring of model outcomes is therefore necessary to ensure that AI-driven marketing supports inclusive participation rather than reinforcing existing gaps. While delivering highly relevant communication can improve engagement, overly personalised marketing may also create concerns regarding surveillance or manipulation. This is particularly relevant for a student-led organisation, where trust and community relationships are central to long-term success. In general, responsible AI use requires maintaining human oversight throughout the marketing process.

Conclusion

The proposed strategy shifts KTHAIS from a traditional, assumption-driven marketing approach to a continuous, AI-driven learning process which allows the organisation to identify evolving patterns of interest, engagement, and intent among students, professionals, and corporate partners. Machine learning methods, including clustering and predictive modelling, enable the identification of relevant audience segments and the estimation of future engagement potential. Through the analysis of digital traces such as website interactions, search behaviour, and online communication, KTHAIS can move from understanding existing audiences towards anticipating future actions, including event participation, membership applications, and partnership opportunities.

NLP enables the transformation of unstructured communication into measurable indicators of sentiment, topics, and interests, providing additional insights into stakeholder needs. Using K-means previously unknown customers can be discovered based on behavioural similarities, while conversion probability and appealing individuals or companies likely to engage with KTHAIS can be predicted using XGBoost. These insights are then integrated with an AI-supported advertising systems, which automatically optimises media buying, advertisement placement, and recommendation timing through techniques such as dynamic creative optimization and real-time bidding.

However, successful implementation requires responsible AI governance. The use of behavioural data must remain transparent, ethical, and aligned with relevant regulations, including GDPR principles. Maintaining human oversight, monitoring potential biases, and protecting stakeholder trust are essential to ensuring that AI supports rather than undermines the values of a student-led non-profit organisation.

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The usage of AI:

AI was used for creating an outline with To-Do items based on the Outline received from the course. For this Gemini Notebook was used. The LLM also received a text containing the report structure from the 'Final Individual Project' page on the course canvas, as well as the authors notes from all 6 lectures attended at that point. The outline received was copy-pasted into the document and gave a tangible place to start. In the end, the AI-printed guideline was not fully used, as the report structure instructions were copy pasted in to guide the writing. The recommended parts to reread from the notes on the lecture were however thoroughly used. The AI was prompted with:

> *Could you give me an outline for a Word document based on the outline in the file "Report Structure". The outline should contain the same rubrics as the report structure, but please add tips of what to read (either articles or what pages from the notes on the Lectures (Lecture _.pdf)?*

Gemini Notebook was also fed al articles that are part to the reference list to help contain a clear overview of what article contained what information. A 'report studio' was then used to output a document each time a report was added containing a clear overview. This way it was simpler for the author to remember from which source what information came. This constantly updated document was helpful when writing in a flow and then not remembering from what source the information originated. The overview often shortened the possible articles to two or three, for which skimming through the articles not took too much time. The following prompt was used when creating the report:

> *Create a comprehensive briefing document that synthesizes the main themes and ideas from the sources. Please do this for each source so it gives an easy overview of which document is which article and that it will be easy to see what information comes from what source. For each source, start with a concise executive summary that presents the most critical takeaways upfront. Then, please provide an examination of the main themes, evidence, and conclusions found in the sources. This analysis should be structured logically with headings and bullet points to ensure clarity. The tone must be objective and incisive. Please write about 500 words for each source (This should thus be done for X sources)*

Lastly ChatGPT was used for spelling checks, a literary review, as well as double checking referencing standards. The output for the spelling check gave a list with incorrectly spelled words that were easily switched using 'find-replace'. Similar was the result for the literary review which gave comments like "*Remove author opinions. This sentence stands out: "The author of the paper believes it to be both.""* which then could be revised and handled. _____ Each part was prompted separately as following:

> *The following text will be a marketing strategy report. The standard language for the text is British English. Please highlight if any American English spellings occur. The text to be corrected follows: "…"*

The following text will be a marketing strategy report. The language should therefor be objective, concise and clear. Please give examples of how to improve the text if issues are found. Especially if the text becomes repetitive, highlight these areas. The text to be corrected follows: "..."

The following text will be a marketing strategy report. Throughout the text APA 7th edition should be used. Furthermore, a total of 27 sources should be used. Please double check that the referencing standard is followed while all sources are referenced. The text to be corrected follows: "..."